



DataCo Supply Chain

Infosys Springboard Virtual Internship

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Introduction

Why Data Preprocessing Matters

Supply Chain Management

Involves coordinating the flow of goods, information, and finances across suppliers, manufacturers, and customers to ensure timely and cost-effective delivery.

Importance of Data Quality

Reliable supply chain decisions depend on clean, accurate, and consistent data. Missing or inconsistent records can distort analysis and mislead conclusions.

Why Preprocessing First

Raw operational data often contains gaps, duplicates, and formatting issues. Preprocessing prepares the dataset so it is trustworthy before any deeper analysis begins.

Objectives

Goals of the preprocessing stage

- 1 Understand the Dataset**
Explore structure, columns, and data types across all 24 fields.
- 2 Identify Missing Values**
Detect gaps across the dataset before they affect analysis.
- 3 Handle Missing Data**
Apply median/mode imputation for numeric and categorical fields.
- 4 Remove Duplicate Records**
Check and eliminate duplicate order entries.
- 5 Standardize the Dataset**
Correct inconsistent category labels and column names.
- 6 Convert Date Columns**
Transform order and shipping dates into datetime format.
- 7 Prepare for Analysis**
Deliver a clean dataset ready for further exploration.



Dataset Overview

2,599

Rows

24

Columns

138

Rows with Missing Values

Dataset: Supply Chain Raw Dataset

Order-level operational data covering shipping performance, customer segments, product categories, sales, and delivery outcomes across five global markets (LATAM, USCA, Pacific Asia, Africa, Europe).

Major Columns

Order ID, Type, Order Status — order identity and payment/status details

Days for shipping (real / scheduled), Delivery Status, Late delivery risk — shipping performance

Category Name, Product Name, Department Name — product classification

Customer Segment, Customer City, Market, Order Region/Country — customer & geography

Sales, Unit Price, Order Item Total, Order Profit Per Order — financial metrics

order date, shipping date, Shipping Mode, Inventory Stock (Units) — logistics detail



Data Preprocessing Steps Using Python

1

Loading & Inspection

Imported Pandas and NumPy, loaded the Excel dataset, reviewed initial records, dimensions (2,599 x 24), and data types for every column.

2

Handling Missing Values

Filled numerical gaps (e.g. Sales, Order Item Total, Inventory Stock) using column median, and categorical gaps (Category Name, Market, Shipping Mode) using column mode.

3

Duplicates & Standardization

Checked for duplicate order records, converted order date and shipping date to datetime format, and standardized inconsistent category labels and column names.

Milestone 1



Python Code Snippets

Key Implementation Lines

1

Loading & Inspection

```
import pandas as pd
import numpy as np

df = pd.read_excel(
    'dataco.xlsx')
print(df.shape)
print(df.dtypes)
```

2

Handling Missing Values

```
df['Sales'].fillna(
    df['Sales'].median(),
    inplace=True)

df['Category Name']
    .fillna(df['Category
    Name'].mode()[0])
```

3

Duplicates & Standardization

```
df.drop_duplicates(
    inplace=True)

df['order date'] =
    pd.to_datetime(
        df['order date'])
```



Preprocessing Outcome

- Missing values across numerical and categorical fields handled successfully
- Duplicate records checked and confirmed absent
- Data types corrected for accurate computation
- Order and shipping dates converted to proper datetime format
- Column names and category labels standardized
- Dataset is now clean, consistent, and ready for further analysis

Conclusion: Preprocessing has transformed the raw DataCo supply chain data into a reliable, well-structured dataset. By resolving missing values, verifying duplicates, and correcting formats, the data is now consistent and analysis-ready, laying a strong foundation for future exploratory and visual insights.



Data Analysis & Visualization

From a clean dataset to visual, decision-ready insights

With preprocessing complete, the cleaned DataCo dataset was used to build an interactive dashboard and perform visual analysis across the core areas of the supply chain.



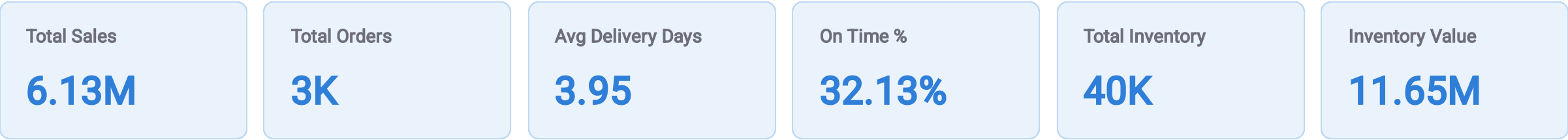
Main Areas Analyzed





Supply Chain Vision: End-to-End Dashboard

A centralized view of sales, orders, delivery performance, inventory, products, categories, regions and cities.



Complete Dashboard Preview



Sales Trend Analysis



What the Visual Shows

Sales performance changes over time.

Periodic increases and decreases can be observed.

The trend provides an overview of sales movement.

Business Relevance

Sales trends help identify periods of stronger or weaker business performance.

Delivery Status Analysis



Categories tracked: Late delivery • Shipping on time • Advance shipping

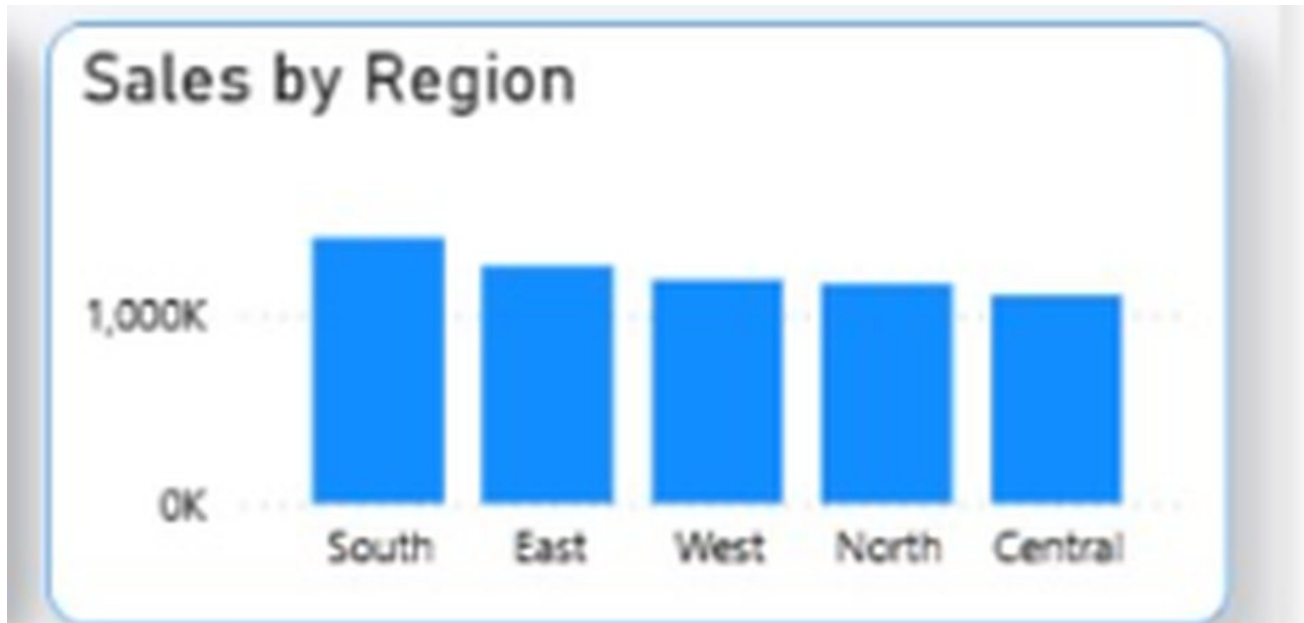
The delivery-status distribution provides a quick view of how orders are progressing through the delivery process.



Key Takeaway

Monitoring delivery status helps identify areas where delivery performance requires attention.

Regional Sales Performance



Regions Compared

South

East

West

North

Central

Key Insight

Regional comparison makes it easier to identify stronger and weaker sales-performing regions.

Regional Delivery Performance



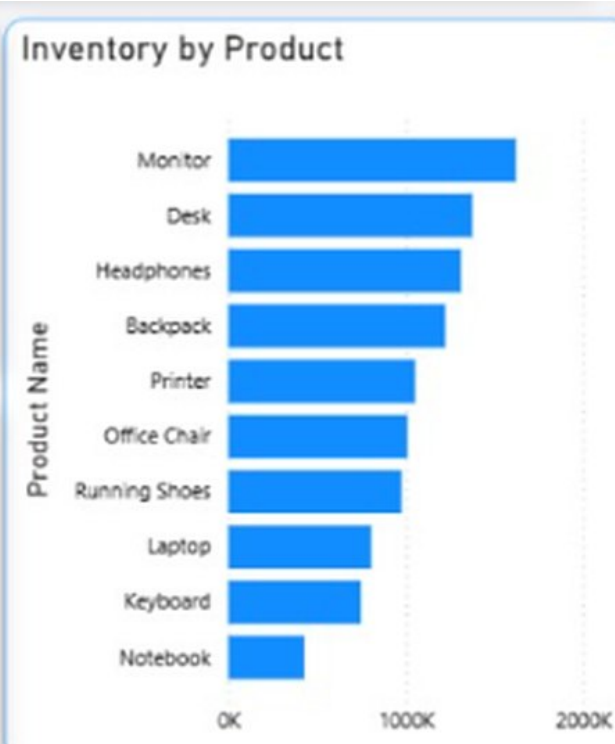
Delivery-status categories: Advance shipping • Late delivery • Shipping on time, compared across South, East, North, Central and West.

Business Relevance

The visualization shows how delivery-status patterns differ across regions. Regional delivery analysis can help management identify locations where delivery performance needs closer monitoring.

Product, Category & City Insights

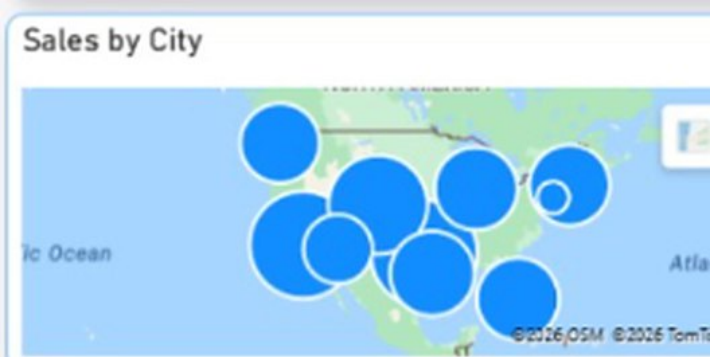
Inventory by Product



Sales by Category



Sales by City



Product-level inventory, category-level sales and city-level distribution provide more detailed views of supply-chain performance.

Key Insights & Conclusion

Total Sales

6.13M

Total Orders

3K

Avg Delivery Days

3.95

On Time %

32.13%

Total Inventory

40K

Inventory Value

11.65M

Analytical Value

- Sales trends provide visibility into performance over time.
- Regional sales analysis highlights geographical differences.
- Delivery-status analysis supports monitoring of shipping performance.
- Product inventory analysis provides visibility into stock levels.
- Category and city analysis provide deeper business insights.

Conclusion

The project transformed the cleaned DataCo supply-chain dataset into a visual analytics solution that provides an end-to-end view of sales, orders, delivery, inventory, products, categories, regions and cities.

“From Data Cleaning to Actionable Supply Chain Insights”

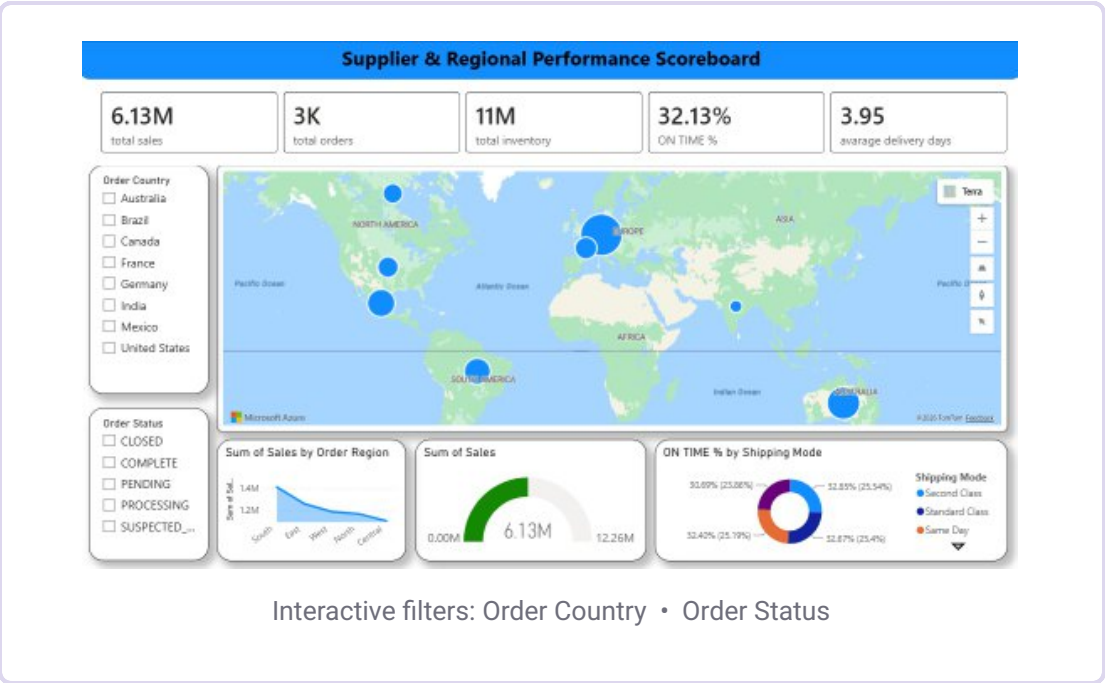
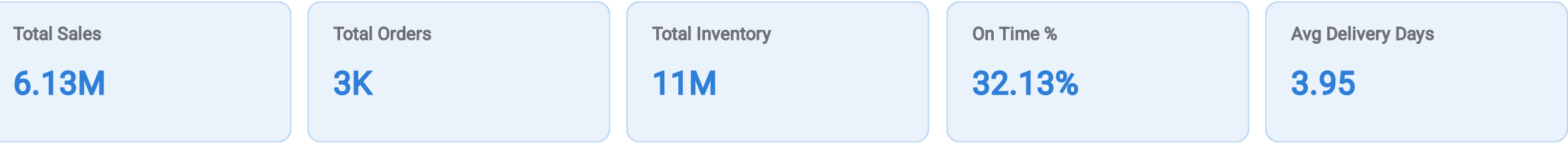
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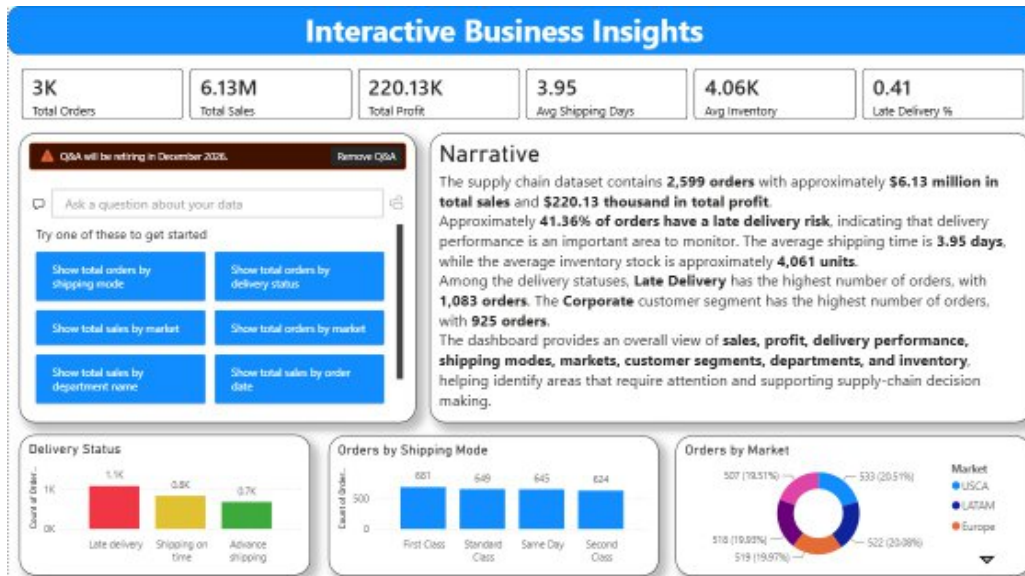
Supplier & Regional Performance Scoreboard

A combined view of sales, orders, inventory and delivery performance across regions.



Transportation Cost & Shipping Efficiency Analysis

No direct transportation-cost field exists in the dataset — efficiency is analyzed via shipping and delivery metrics.



Shipping mode, delivery status and market breakdowns from the cleaned dataset

Metrics Analyzed

- Shipping Mode
- Actual vs. Scheduled Shipping Days
- Delivery Status
- Late Delivery Risk
- Order Region • Order Country

Key Questions

- Which shipping modes are more efficient?
- Which regions experience greater delivery risk?
- How does shipping performance vary by delivery status?
- Which logistics areas require improvement?

Business Insight

Shipping efficiency influences delivery reliability and operational performance. Comparing modes, regions and outcomes reveals opportunities for logistics optimization.



Route & Carrier Performance Analysis

A drill-down view of routes, departments, shipping modes and delivery outcomes across markets.

Total Sales

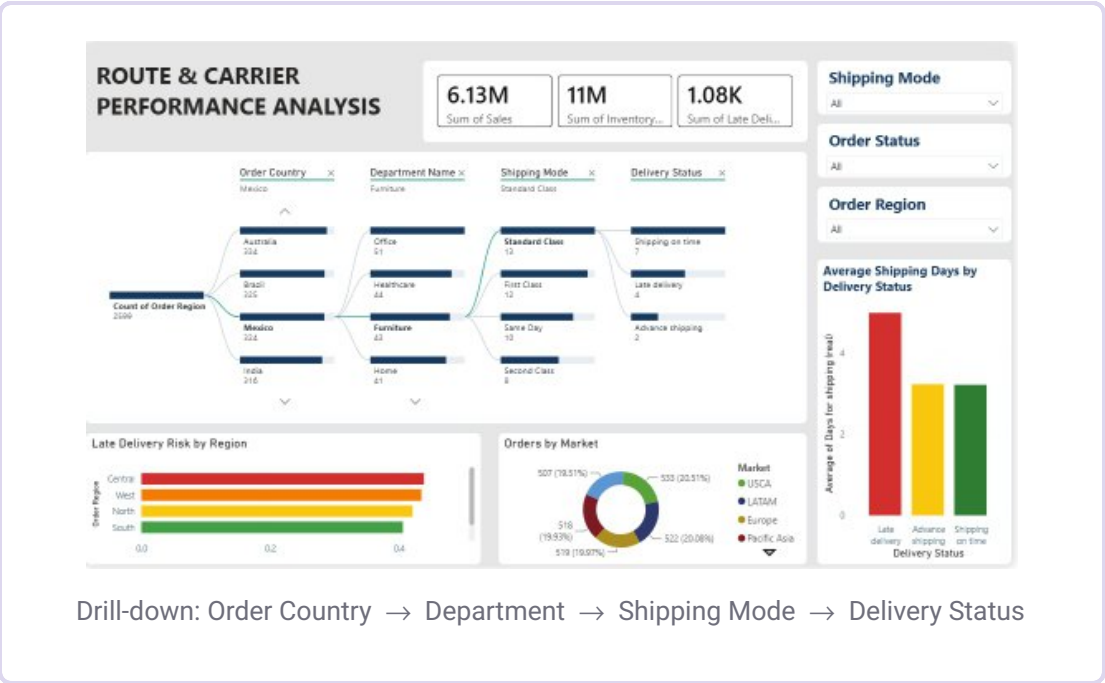
6.13M

Inventory Value

11M

Late Delivery Risk

1.08K



Key Insights

Route Visibility — country, department, shipping mode and delivery combinations can be investigated.

Delivery Risk — regional late-delivery risk can be identified and monitored.

Shipping Efficiency — average shipping days can be compared across delivery statuses.

Market Coverage — order distribution across markets provides geographic context.

Operational Action — high-risk regions and route combinations can be prioritized.

Note: carrier-level performance is not claimed as best or worst unless actual carrier data is present in the source.



Advanced KPI Visualization

Comparing delivery performance across region, customer segment and shipping mode.

Total Orders

3K

Total Sales

6.13M

Avg Delivery Days

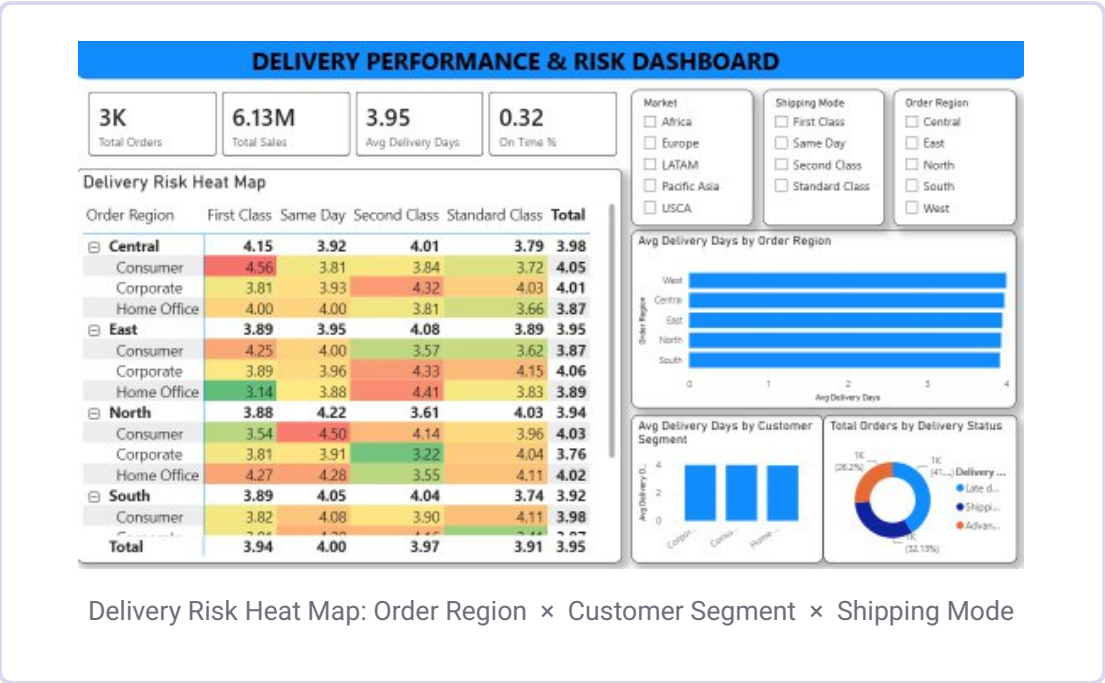
3.95

On Time %

32.13%

Late Delivery Risk

~1.08K



Delivery Risk Heat Map: Order Region × Customer Segment × Shipping Mode

Heat Map Interpretation

The heat map compares delivery performance across region, customer segment and shipping mode, helping identify combinations linked to higher delivery time or risk.

KPI Insight Cards

- Delivery Speed** — average delivery time is approximately 3.95 days.
- On-Time Performance** — approximately 32.13% of orders are on time.
- Delivery Risk** — late-delivery risk should be monitored by region and mode.
- Segment Performance** — customer segments show different delivery patterns.



SALES & ORDER ANALYSIS

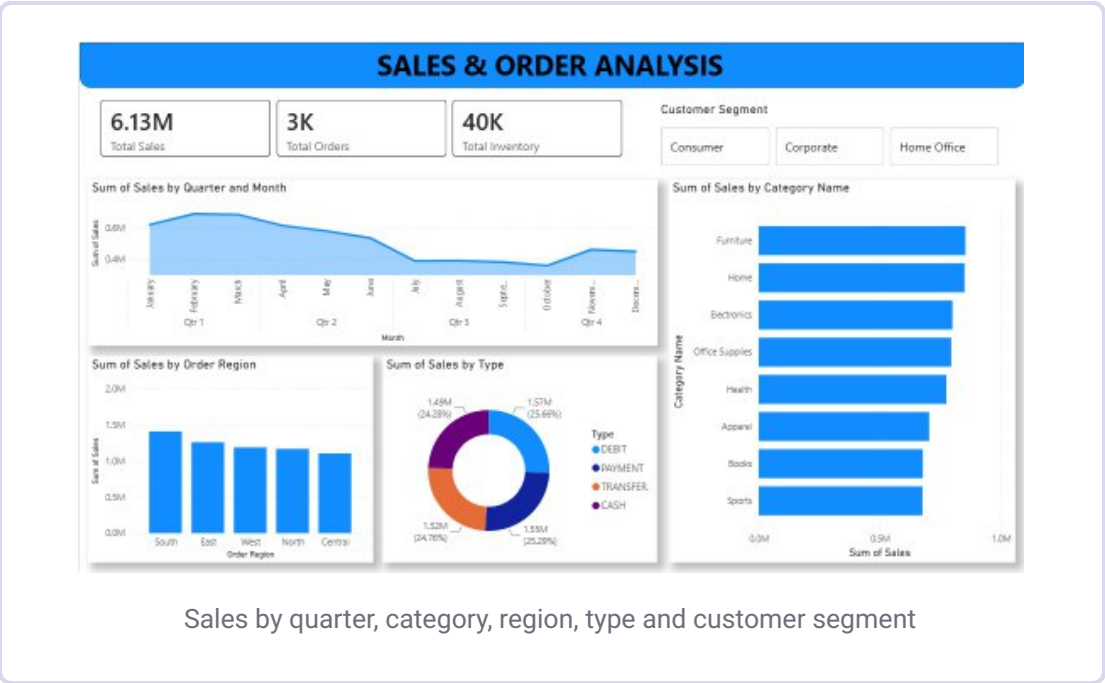
Revenue, order volume and customer-segment performance from the cleaned DataCo dataset.

Total Sales
6.13M

Total Orders
3K

Total Inventory
40K

Customer Segment
Consumer | Corporate | Home Office



Key Insights

- Sales Trend** — performance varies across months and quarters.
- Regional Strength** — South is the strongest displayed region by sales.
- Category Mix** — Furniture and Home are among the leading displayed categories.
- Payment Types** — sales are distributed across Debit, Payment, Transfer and Cash.
- Customer Segment** — filters allow comparison of Consumer, Corporate and Home Office performance.

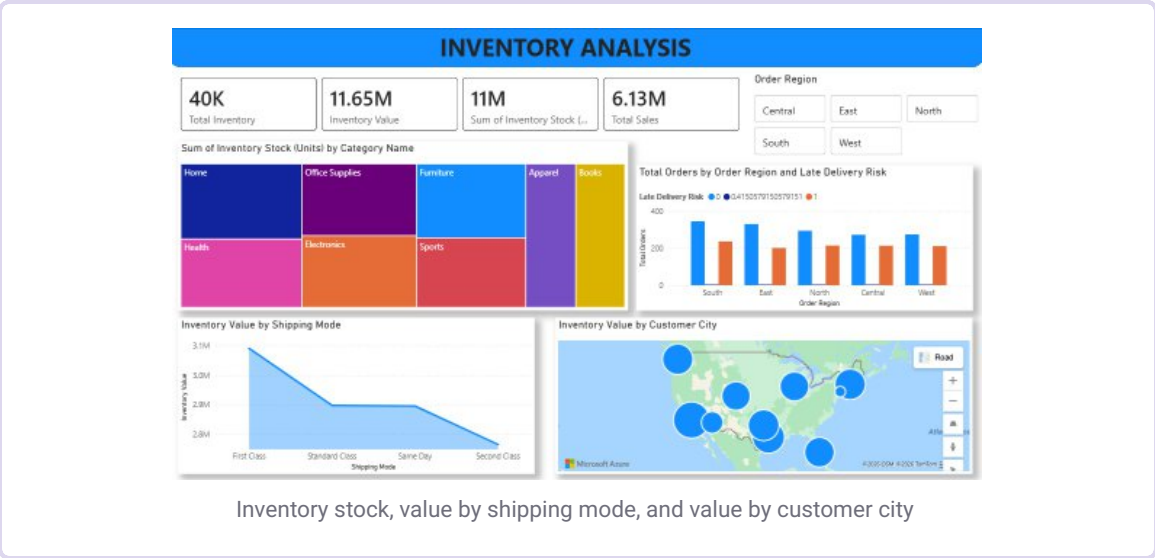
Business Value

Sales and order analysis provides visibility into revenue trends, regional performance, category contribution and customer-segment behavior, supporting data-driven business decisions.



INVENTORY & DELIVERY PERFORMANCE ANALYSIS

Combining inventory position with delivery and shipping performance across regions and shipping modes.



Inventory Insights

- Inventory spans multiple product categories (~40K units).
- Inventory value is approximately 11.65M.
- Inventory value varies by shipping mode and customer city.



Delivery Insights

- Average delivery time is approximately 3.95 days.
- On-time delivery is approximately 32.13% — a key improvement area.
- Late-delivery risk varies across order regions.

Business Value: combining inventory and delivery analysis helps management monitor stock position while identifying logistics areas that need attention.

Key Insights & Conclusion

Total Sales	Total Orders	Avg Delivery Days	On-Time Delivery	Total Inventory	Inventory Value
6.13M	3K	3.95	32.13%	40K	11.65M

Key Insights

- Sales Performance** — visibility into sales trends across time, regions, categories and order types.
- Regional Performance** — South is the strongest displayed sales region; regional comparisons highlight differences in business performance.
- Inventory Visibility** — approximately 40K inventory units and 11.65M inventory value show the importance of effective monitoring.
- Delivery Performance** — average delivery time is approximately 3.95 days and on-time delivery is approximately 32.13%, an important improvement area.
- Logistics & Risk** — shipping-mode, delivery-status and regional late-delivery-risk analysis adds visibility into logistics performance.
- Interactive Decision Support** — the Power BI dashboards allow users to explore sales, inventory, delivery, regions, categories and other supply-chain dimensions interactively.

Conclusion

The project transformed the cleaned DataCo supply-chain dataset into an interactive visual analytics solution covering sales, orders, inventory, delivery performance, shipping modes, regions, categories and operational risk. The dashboards provide a decision-support framework that helps identify performance gaps, monitor key supply-chain KPIs and prioritize opportunities for operational improvement.

“From Data Cleaning to Actionable Supply Chain Insights”

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DataCo Supply Chain Analytics

Milestone 4 – Interactive Dashboard Development

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Four Dashboards for End-to-End Supply Chain Analysis

1

Executive Overview

Overall supply-chain performance and management KPIs

2

Warehouse Efficiency

Orders, shipping efficiency and delivery performance

3

Inventory & Product Performance

Inventory, products, sales and profitability

4

Delivery & Regional Performance

Delivery reliability, markets, countries, regions and shipping modes

Together, the four dashboards provide an integrated view of supply-chain performance — from management KPIs to operational and regional analysis.

Executive Overview



Key KPIs

Business Value: Enables management to quickly monitor overall supply-chain performance and identify areas requiring attention.

3K

Total Orders

6.13M

Total Sales

220.13K

Total Profit

3.95

Avg Shipping Days

Warehouse Efficiency

Key KPIs

Business Value: Helps identify operational bottlenecks and opportunities to improve warehouse and shipping efficiency.

3K

Total Orders

3.95

Avg Shipping Days

42%

Late Delivery Rate



Inventory & Product Performance



Key KPIs

Business Value: Provides visibility into inventory position, product performance, category contribution and profitability.

10
Total Products

4.06K
Avg Inventory

6.13M
Total Sales

14K
Total Quantity

Delivery & Regional Performance

Key KPIs

Key Insight: On-time delivery at ~32% flags delivery reliability as a critical improvement area. Performance varies significantly across markets and regions.

3K

Total Orders

41.7%

Late Delivery Rate

32.1%

On-Time Delivery

0.45

Avg Delivery Gap



From Data to Actionable Supply Chain Insights

The four interactive dashboards transform supply-chain data into a decision-support solution covering overall performance, warehouse efficiency, inventory and product performance, and delivery and regional performance.

1

Data

Raw supply-chain records

2

Analysis

KPI extraction & modelling

3

Dashboards

4 interactive Power BI views

4

Insights

Actionable business decisions